Asynchronous communications (MQTT)

What is MQTT?

Message Queuing Telemetry Transport (MQTT)

A lightweight publish/subscribe messaging protocol designed for:

- constrained devices
- unreliable or low-bandwidth networks
- IoT and edge computing scenarios

👉 Unlike HTTP (request/response), MQTT is **event-driven and asynchronous**.

Key Characteristics

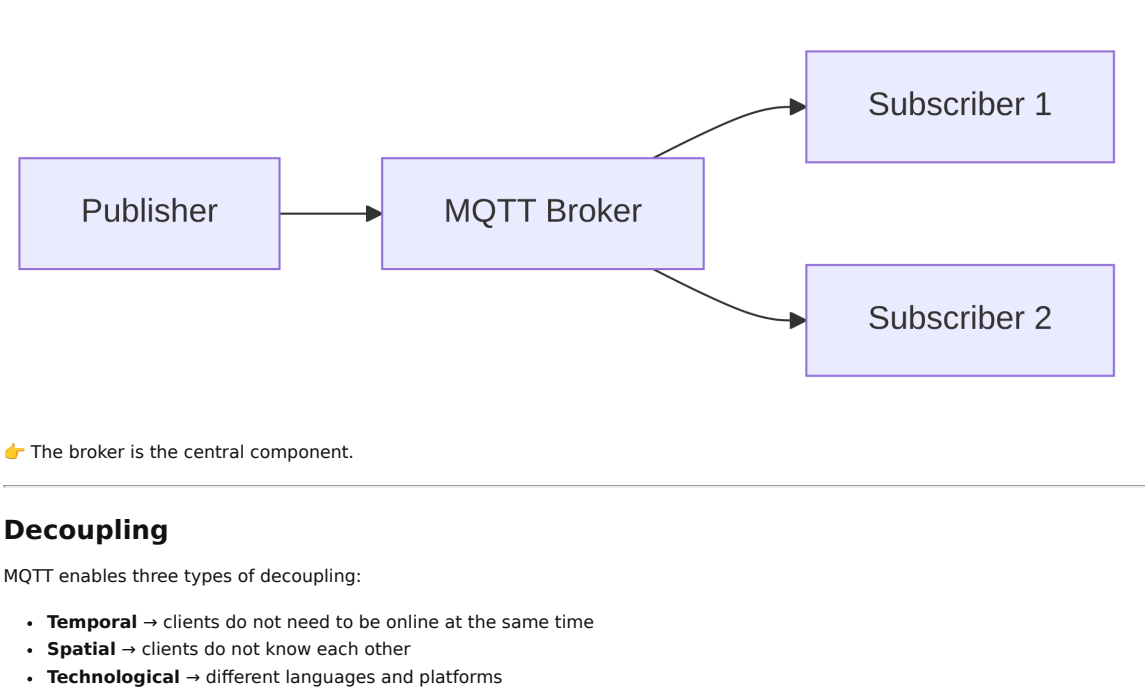
- Low bandwidth usage
- Low latency
- Asynchronous communication
- Decoupled architecture
- Native fit for IoT

👉 Ideal when you have many distributed producers and consumers.

MQTT Architecture

Core components:

- **Publisher** → sends messages
- **Broker** → routes messages
- **Subscriber** → receives messages


Publisher
MQTT Broker
Subscriber 1
Subscriber 2

👉 The broker is the central component.

Decoupling

MQTT enables three types of decoupling:

- **Temporal** → clients do not need to be online at the same time
- **Spatial** → clients do not know each other
- **Technological** → different languages and platforms

👉 This is why MQTT works well with microservices.

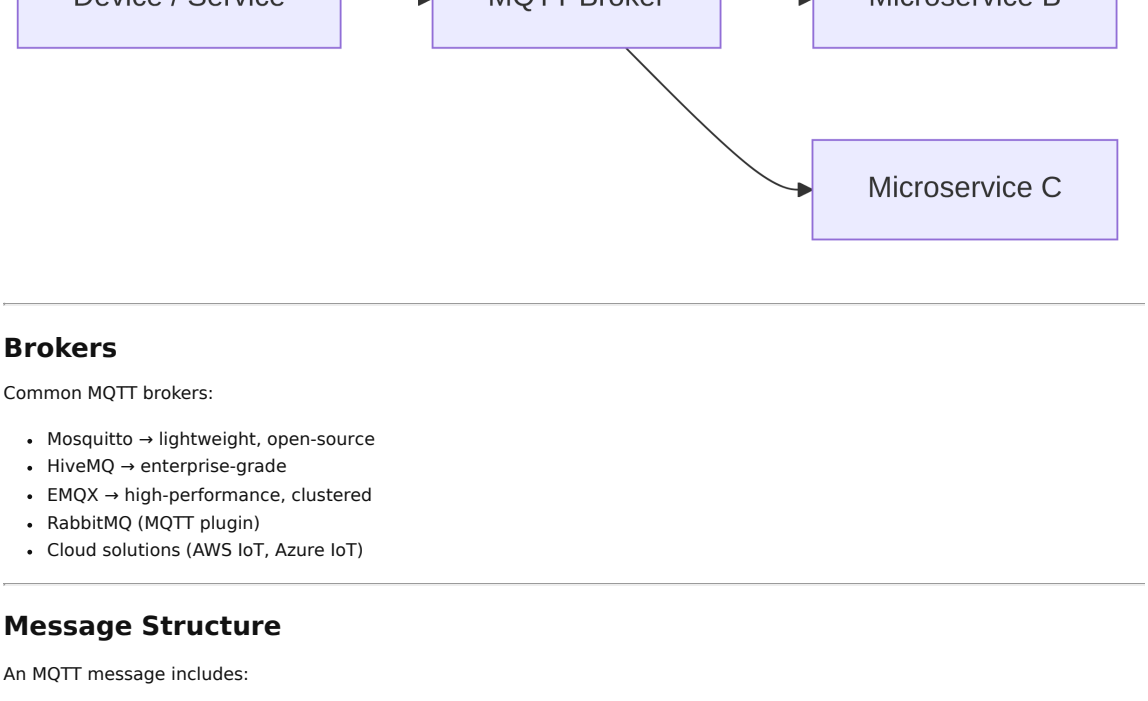
MQTT in Microservices

MQTT acts as a:

👉 **Lightweight Event Bus**

Typical use cases:

- notifications
- telemetry
- real-time updates


Device / Service
MQTT Broker
Microservice A
Microservice B
Microservice C

Brokers

Common MQTT brokers:

- Mosquitto → lightweight, open-source
- HiveMQ → enterprise-grade
- EMQX → high-performance, clustered
- RabbitMQ (MQTT plugin)
- Cloud solutions (AWS IoT, Azure IoT)

Message Structure

An MQTT message includes:

- **Topic** → routing key
- **Payload** → data (JSON, binary, etc.)
- **QoS** → delivery guarantee
- **Retain flag**
- **Properties** (MQTT 5.0)

Topics

Topics are hierarchical strings:

```
bank/account/123/deposit
sensors/temperature/lab1
```

👉 Topics define routing, not the payload.

Topic Design (Best Practices)

Recommended pattern:

```
<domain>/<entity>/<id>/<event>
```

Examples:

- iot/device/thermostat-1/temperature
- bank/account/123/withdraw

❗ Avoid:

- generic topics like `data`, `events`
- encoding routing logic inside payloads

Subscriptions & Wildcards

- `+` → matches one level
- `#` → matches multiple levels

Examples:

```
home/+/temperature
home/#
```

Quality of Service (QoS)

- QoS 0 → at most once
- QoS 1 → at least once
- QoS 2 → exactly once

QoS Trade-offs

QoS	Guarantee	Cost	Typical Use
0	None	Low	High-frequency sensors
1	At least once	Medium	Business events
2	Exactly once	High	Critical operations

👉 QoS 2 is rarely used in practice due to overhead.

👉 QoS 1 may produce duplicates

- idempotent consumers
- message IDs
- deduplication logic

Retained Messages vs Event Messages (MQTT)

Retained Messages

- The broker stores the **last message per topic**
- New subscribers immediately receive the **latest retained value**

👉 This is used to represent the **current state** of a system

Typical use case:

- device status (online/offline)
- current configuration
- latest known sensor value

Event Messages (Non-Retained)

- Messages are **transient**
- Delivered only at the time of publication
- Not stored by the broker

👉 This represents a **change or event over time**

Typical use case:

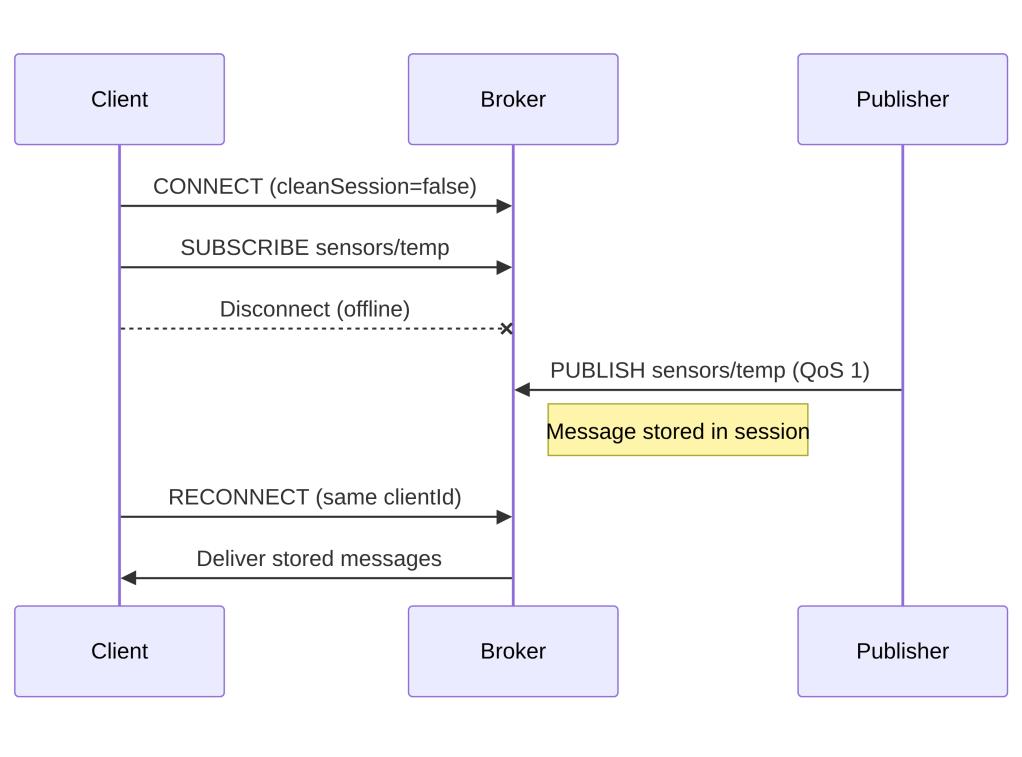
- temperature readings
- alerts
- activity logs

Key Insight

👉 Retained messages represent "what is true now"
👉 Event messages represent "what just happened"

Last Will and Testament (LWT)

- Defined when the client connects
- Published by the broker on unexpected disconnect


Client
disconnect failure
Broker
Subscribers receive LWT

Use case:

- failure detection in distributed systems

Persistent Sessions (MQTT)

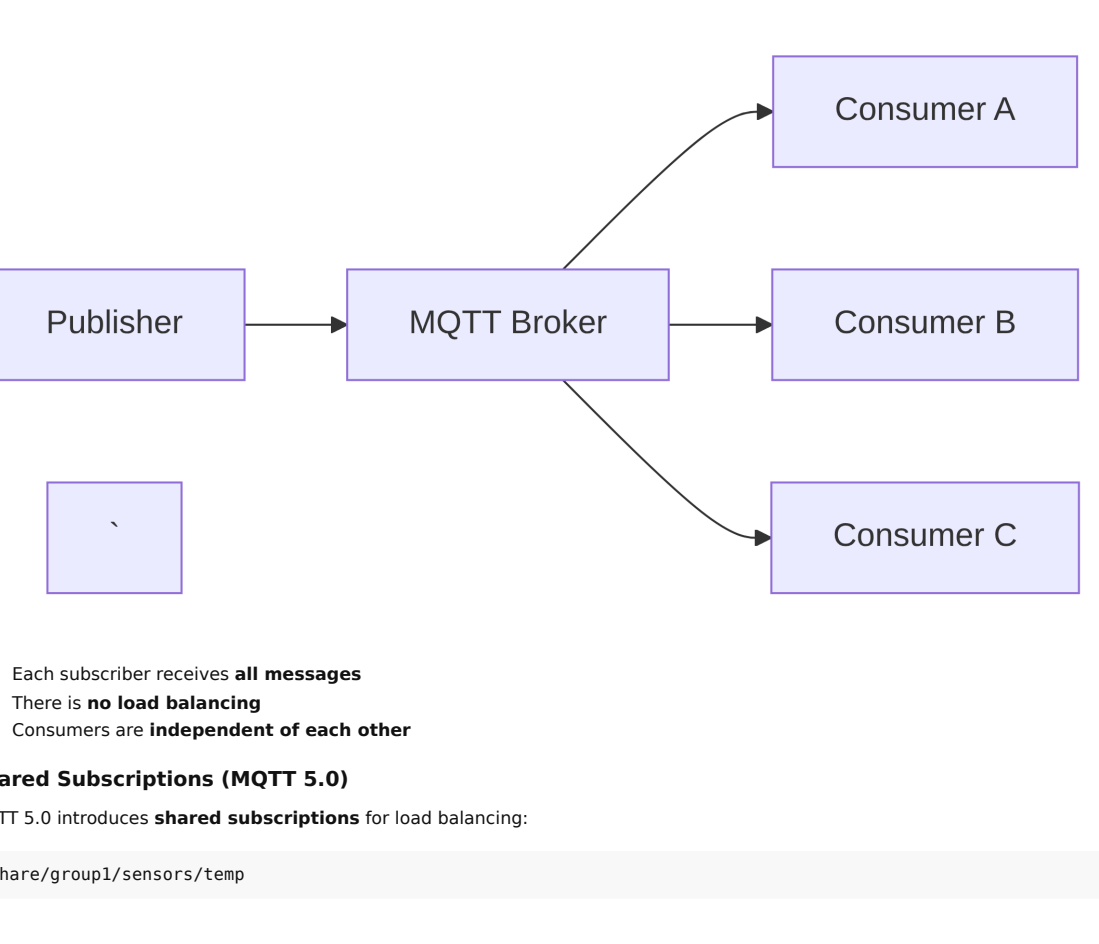
clean session = false

When a client reconnects, the broker preserves its session state instead of resetting everything.

The broker stores:

- subscriptions
- undelivered QoS 1/2 messages
- in-flight messages (awaiting acknowledgment)

👉 Useful for intermittently connected devices (IoT, mobile, edge systems)

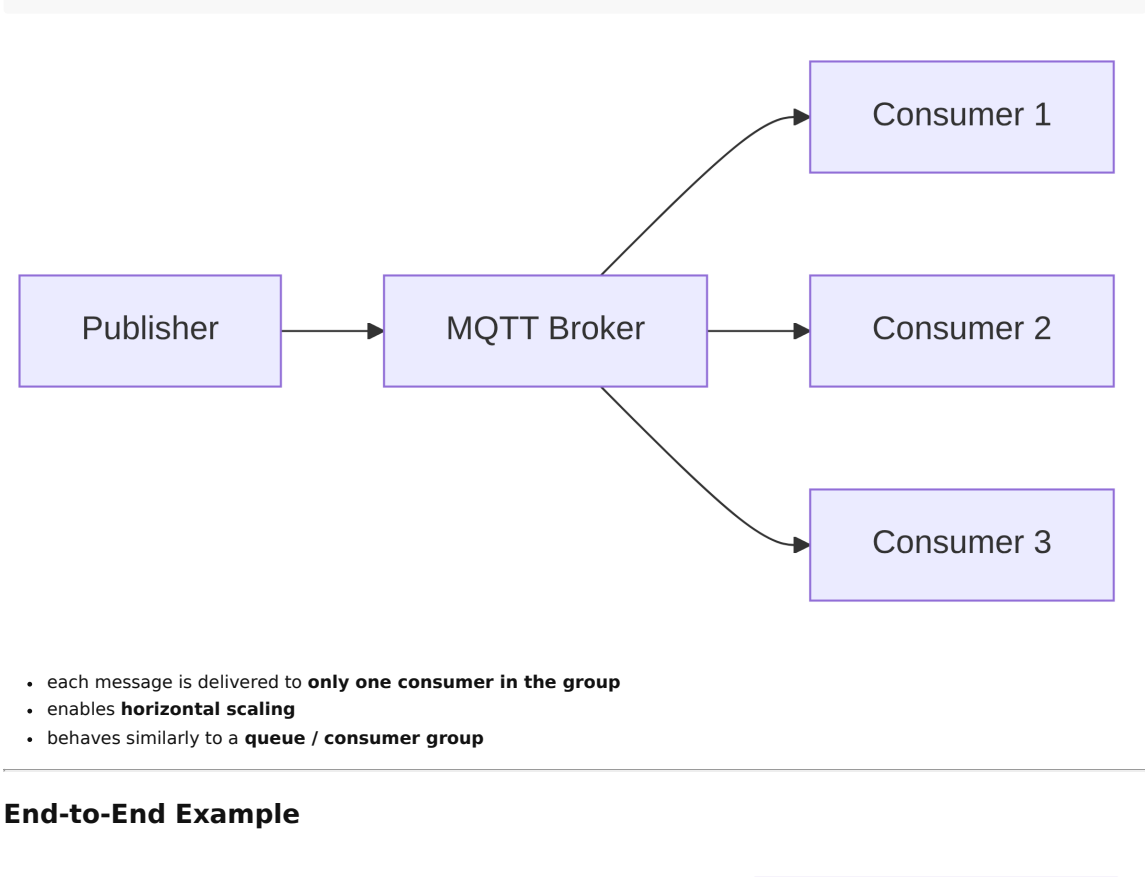

Client
Broker
Publisher
CONNECT (cleanSession=false)
SUBSCRIBE sensors/temp
Disconnect (offline)
PUBLISH sensors/temp (QoS 1)
Message stored in session
RECONNECT (same clientId)
Deliver stored messages
Client
Broker
Publisher

Multiple Consumers on the Same MQTT Topic

Default Behavior (Standard MQTT)

In MQTT, when multiple clients subscribe to the **same topic**, each of them receives a **full copy** of every message.

👉 This is a **publish/subscribe broadcast model (fan-out)**

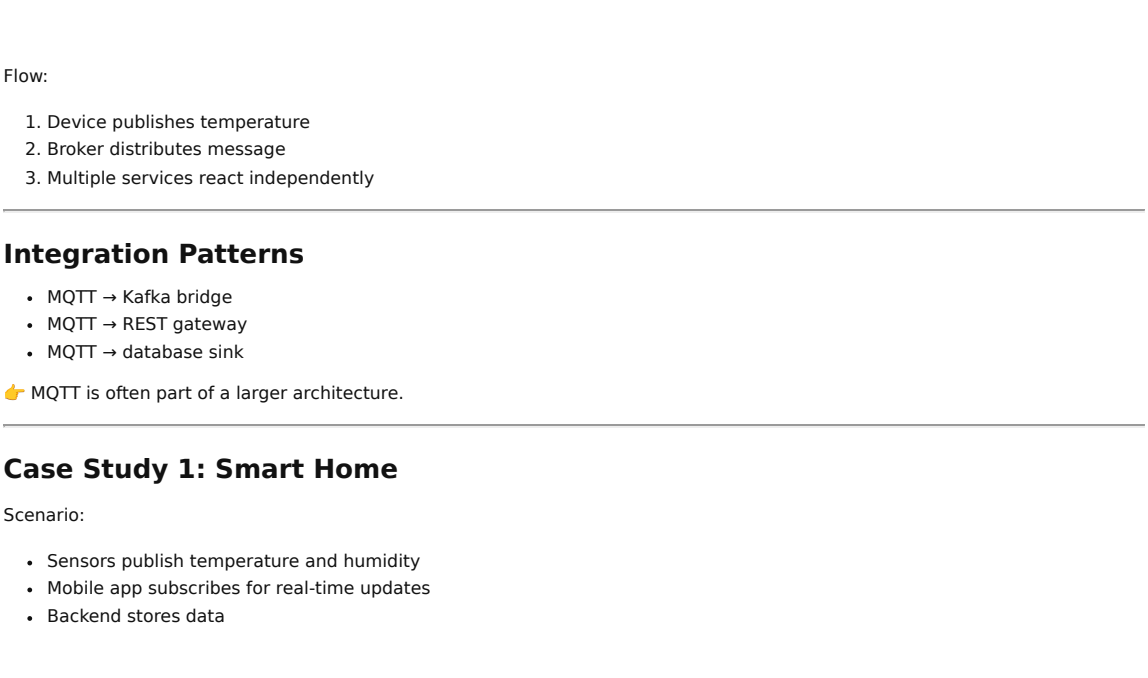

Publisher
MQTT Broker
Consumer A
Consumer B
Consumer C
`

- Each subscriber receives **all messages**
- There is **no load balancing**
- Consumers are **independent of each other**

Shared Subscriptions (MQTT 5.0)

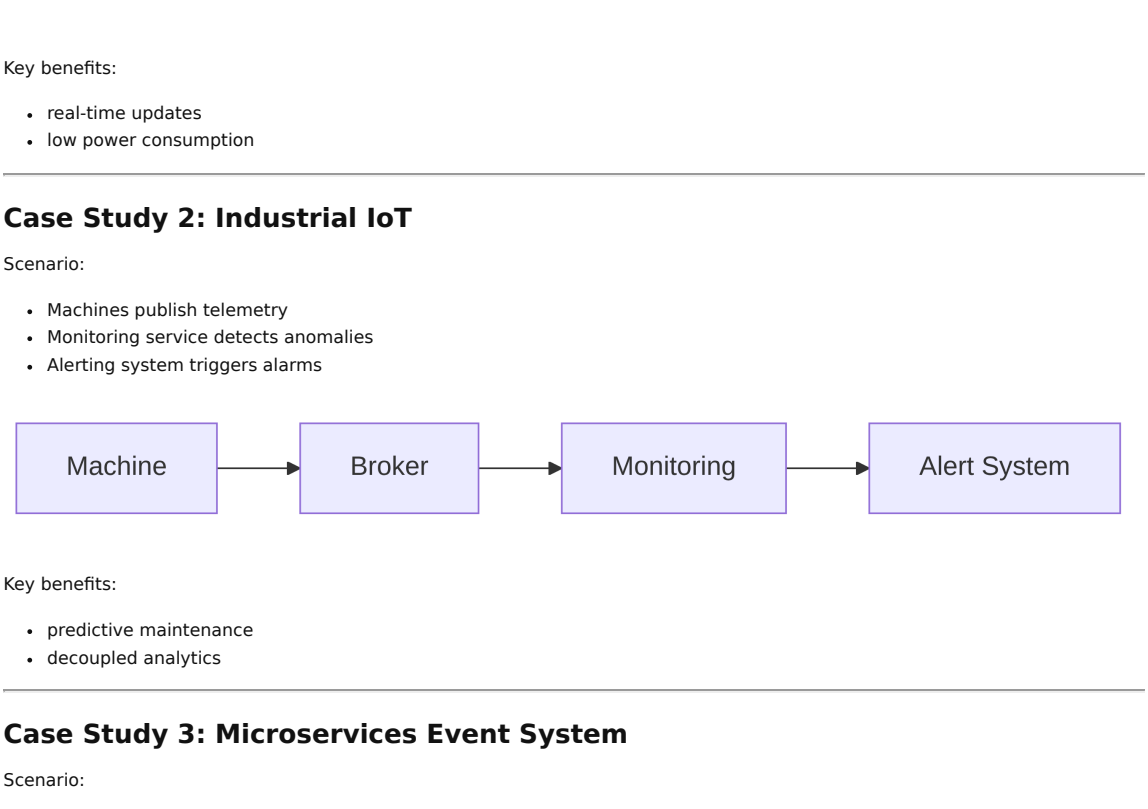
MQTT 5.0 introduces **shared subscriptions** for load balancing:

```
$share/group1/sensors/temp
```


Publisher
MQTT Broker
Consumer 1
Consumer 2
Consumer 3

- each message is delivered to **only one consumer in the group**
- enables **horizontal scaling**
- behaves similarly to a **queue / consumer group**

End-to-End Example


IoT Device
MQTT Broker
Monitoring Service
Alerting Service
Storage Service

Flow:

1. Device publishes temperature
2. Broker distributes message
3. Multiple services react independently

Integration Patterns

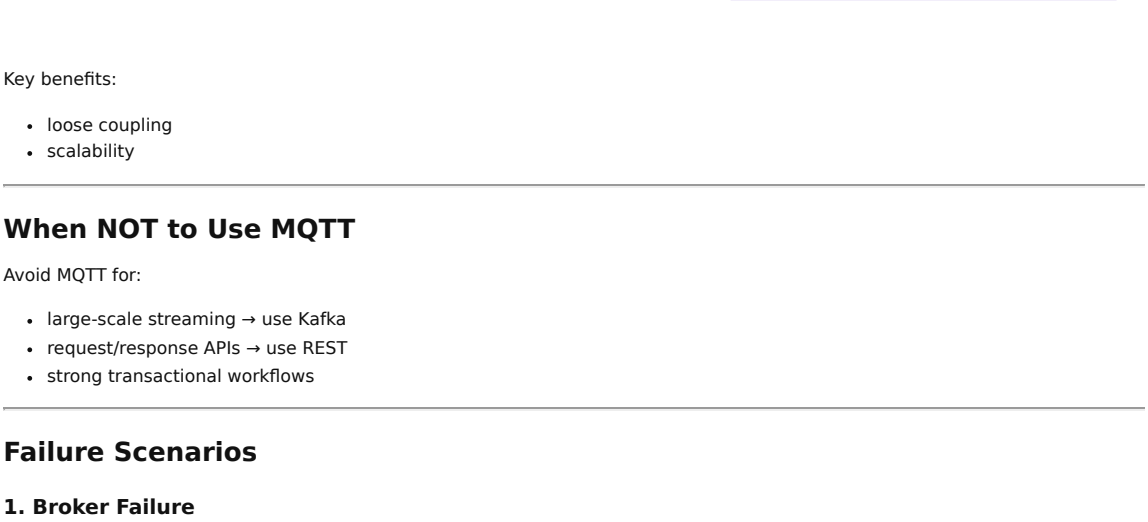
- MQTT → Kafka bridge
- MQTT → REST gateway
- MQTT → database sink

👉 MQTT is often part of a larger architecture.

Case Study 1: Smart Home

Scenario:

- Sensors publish temperature and humidity
- Mobile app subscribes for real-time updates
- Backend stores data


Sensor
Broker
Mobile App
Backend Storage

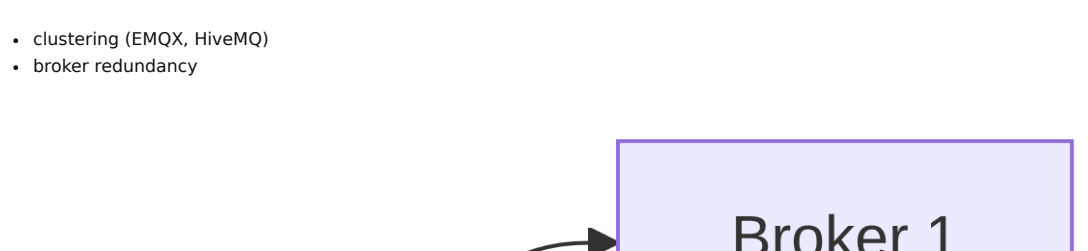
Key benefits:

- real-time updates
- low power consumption

Case Study 2: Industrial IoT

Scenario:

- Machines publish telemetry
- Monitoring service detects anomalies
- Alerting system triggers alarms


Machine
Broker
Monitoring
Alert System

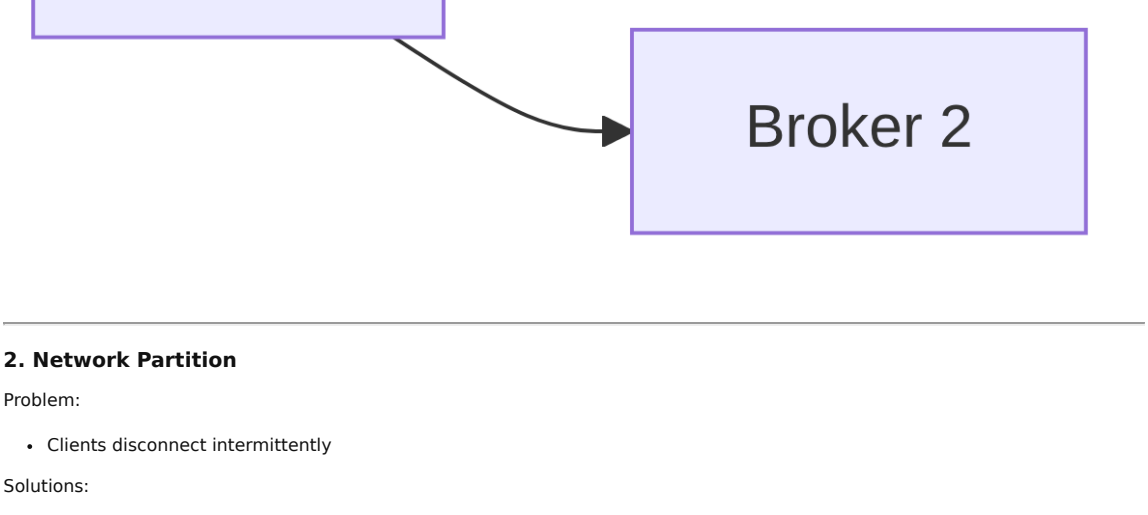
Key benefits:

- predictive maintenance
- decoupled analytics

Case Study 3: Microservices Event System

Scenario:

- Order service publishes events
- Payment, shipping, and notification services react


Order Service
Broker
Payment Service
Shipping Service
Notification Service

Key benefits:

- loose coupling
- scalability

When NOT to Use MQTT

Avoid MQTT for:

- large-scale streaming → use Kafka
- request/response APIs → use REST
- strong transactional workflows

Failure Scenarios

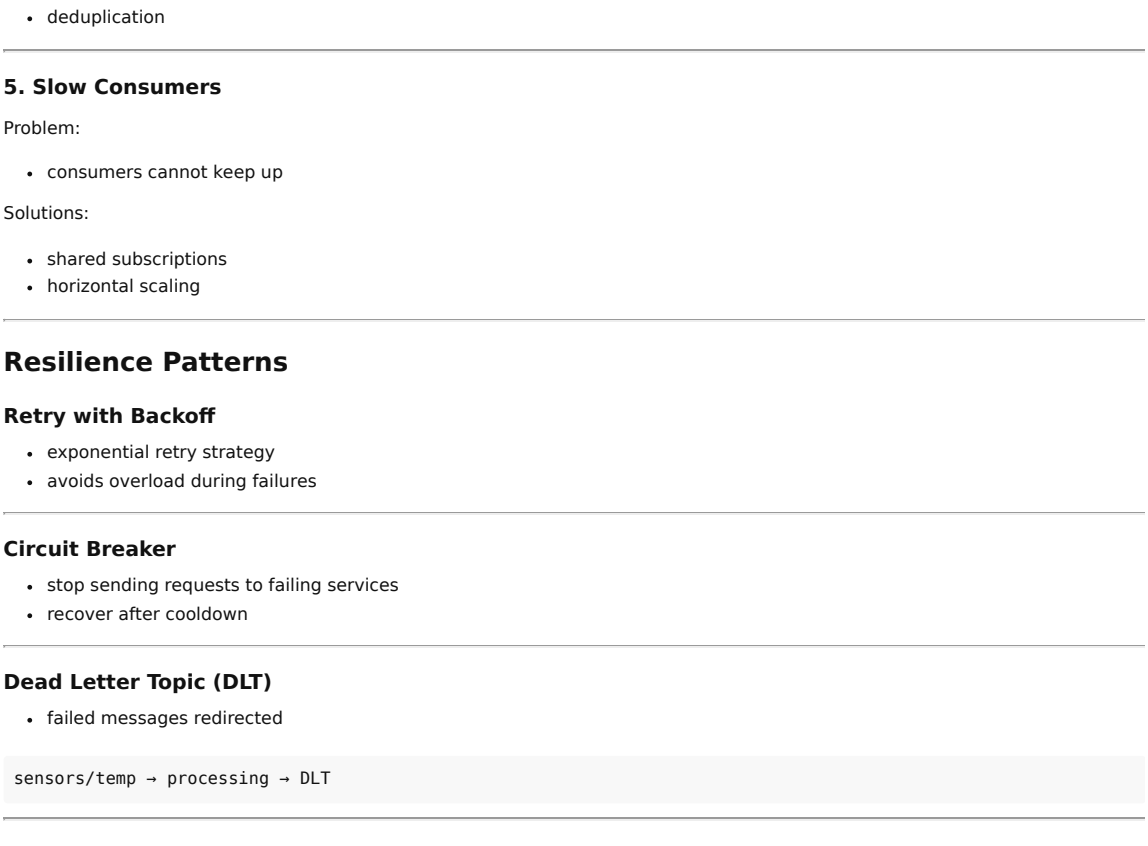
1. Broker Failure

Problem:

- Single broker becomes a bottleneck or point of failure

Solution:

- clustering (EMQX, HiveMQ)
- broker redundancy


Clients
Broker 1
Broker 2

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2. Network Partition

Problem:

- Clients disconnect intermittently

Solutions:

- persistent sessions
- message buffering
- retries with backoff

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3. Message Loss (QoS 0)

Problem:

- messages can be dropped

Solution:

- use QoS 1 or QoS 2
- implement retry logic

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4. Duplicate Messages (QoS 1)

Problem:

- at-least-once delivery → duplicates

Solution:

- idempotent consumers
- deduplication

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5. Slow Consumers

Problem:

- consumers cannot keep up

Solutions:

- shared subscriptions
- horizontal scaling

Resilience Patterns

Retry with Backoff

- exponential retry strategy
- avoids overload during failures

Circuit Breaker

- stop sending requests to failing services
- recover after cooldown

Dead Letter Topic (DLT)

- failed messages redirected

```
sensors/temp → processing → DLT
```

Idempotent Consumer

- process each message once logically

Resources

- MQTT Official Specification
- Eclipse Mosquitto
- HiveMQ Documentation
- MQTT Essentials Series